

Course: BUS123 — Solving Business Problems with Technology

Track: MATH · Module 07 · Lesson 01

Est. Read: 25–30 min

Connecting to Prior Learning

In the previous module, you learned how businesses set selling prices — applying trade discounts to reduce the retail price before quoting a buyer, and marking up costs to establish a profitable price point. You built those calculations in Excel using cell references so that any change in an input automatically updated the entire chain.

This lesson adds a critical layer: **government taxes**. Once a price is set, taxes are applied at specific points in a defined order — and getting that order wrong produces incorrect totals, incorrect revenue reporting, and potential legal liability. Today we examine three distinct tax types — sales, excise, and property — each of which places a different obligation on the business.

Part 1 — Sales Tax

What Is Sales Tax?

Sales tax is a broad-based consumption tax levied by state governments on the retail sale of goods and certain services. Every state sets its own rate; some municipalities add a local surcharge on top. As of 2024, state rates range from 0% (Oregon, Montana, New Hampshire, Delaware, Alaska) to over 7% before local additions.

■ **The Merchant's Role:** When a business collects sales tax from a customer, those funds are NOT revenue. They are a liability held in trust for the state government. The business is acting as a collection agent. Misclassifying collected tax as earned revenue is a serious accounting error.

The Order of Operations

The calculation sequence is strictly defined. Trade discounts reduce the taxable base — they must come **before** tax is calculated. Shipping charges are added **after** tax has been computed. Reversing these steps produces an incorrect tax figure.

Step	Operation	Formula
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1	Subtract trade discount from retail price	$\text{Taxable} = \text{Price} - \text{Discount}$
2	Apply tax rate to taxable subtotal	$\text{Tax} = \text{Taxable} \times \text{Rate}$
3	Add shipping to get total price	$\text{Total} = \text{Taxable} + \text{Tax} + \text{Shipping}$

Worked Example — Tidal Goods Co.

Tidal Goods Co. sells a bulk supply order to a wholesale partner. Retail price: **\$480.00**. Trade discount for the partner: **\$80.00**. Local sales tax rate: **6%**. Flat shipping: **\$15.00**.

Item	Amount
Retail Price	\$480.00
Less: Trade Discount	– \$80.00
Taxable Subtotal	\$400.00
× Sales Tax Rate (6%)	× 0.06
Tax Owed	\$24.00
Taxable Subtotal + Tax	\$424.00
+ Shipping	+ \$15.00
Total Price	\$439.00

Back-Calculating Actual Sales

When a retailer's register total includes both sales revenue and collected tax, the actual pre-tax revenue must be isolated using the reverse formula:

Actual Sales	$= \text{Total Receipts} / (1 + \text{Tax Rate})$
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Example: Tidal Goods Co.'s register shows \$52,000 in total daily receipts at a 6% tax rate. Actual sales = $\$52,000 \div 1.06 =$ **\$49,056.60**. The remaining \$2,943.40 belongs to the state.

Part 2 — Excise Tax

What Is Excise Tax?

Excise tax is a targeted, selective tax applied to specific categories of goods considered luxuries or nonessentials — fuel, alcohol, tobacco, certain appliances, firearms, and air travel. Unlike sales tax, which is collected at the retail register, excise tax is typically paid **upstream** by manufacturers or wholesalers. By the time the product reaches a consumer, the excise tax is already embedded in the sticker price.

■ Strategic Impact: Because excise tax raises the cost of acquiring inventory, it directly compresses gross margin. Managers must evaluate demand elasticity — whether the market will absorb a price increase — before passing the cost to customers.

When Both Taxes Apply — Never Compound Them

When both sales tax and excise tax apply to the same item, each is calculated independently from the original base cost. The rates are **not** compounded (do not apply the second tax to the already-taxed amount).

Worked Example — Meridian Advisory Group purchases a commercial appliance for \$6,200. Sales tax: 7%. Excise tax: 8%.

Tax Type	Calculation	Amount
Base Cost	Given	\$6,200.00
Sales Tax (7%)	$\$6,200 \times 0.07$	\$434.00
Excise Tax (8%)	$\$6,200 \times 0.08$	\$496.00
Total Cost	$\$6,200 + \$434 + \$496$	\$7,130.00

■ Shortcut: Add the two rates first ($7\% + 8\% = 15\%$), then multiply: $\$6,200 \times 1.15 = \$7,130$. This works because both rates are applied to the same base — they can be combined before multiplying.

Part 3 — Property Tax

What Is Property Tax?

Property tax is collected by local municipalities to fund community services: schools, roads, fire and police protection, public libraries, and infrastructure. It applies to both **real property** (land and buildings) and **business personal property** (movable assets such as computers, furniture, and machinery).

Municipalities do not tax property at its full market value. Instead, they apply an **assessment rate** to calculate a lower **assessed value**, which is the taxable base.

■ Triple Net Lease (NNN) Warning: In a NNN lease, the tenant — not the landlord — pays property taxes, building insurance, and maintenance. This makes the tenant's effective rent variable, tied directly to the municipality's annual budget. A town that approves a larger budget raises every NNN tenant's costs automatically.

The Three-Step Calculation

Step	What You Calculate	Formula
1	Assessed Value	$\text{Assessment Rate} \times \text{Market Value}$
2	Tax Rate — round to 4 decimal places	$\text{ROUND}(\text{Budget} / \text{Total AV}, 4)$
3	Annual Tax Due	$\text{Tax Rate} \times \text{Assessed Value}$

Worked Example — Meridian Advisory Group

Given: Market Value \$820,000 · Assessment Rate 40% · Town Budget \$310,000 · Total Assessed Value in Town \$4,750,000

Step	Calculation	Result
1 — Assessed Value	$0.40 \times \$820,000$	\$328,000.00
2 — Tax Rate (raw)	$\$310,000 / \$4,750,000$	0.0652631...
2 — Tax Rate (rounded)	$=\text{ROUND}(\text{rate}, 4)$	0.0653
3 — Annual Tax Due	$0.0653 \times \$328,000$	\$21,418.40

Formula Reference

Formula Name	Algebraic / Excel Form
Sales Tax — Taxable Subtotal	$= \text{Price} - \text{Trade Discount}$
Sales Tax — Tax Owed	$= \text{Taxable} \times \text{Tax Rate}$

Sales Tax — Total Price	= Taxable + Tax + Shipping
Back-Calculate Actual Sales	= Total Receipts / (1 + Tax Rate)
Excise — Each tax separately	= Base × SalesRate ; = Base × ExciseRate
Excise — Combined shortcut	= Base × (1 + SalesRate + ExciseRate)
Property — Assessed Value	= Assessment Rate × Market Value
Property — Tax Rate (4 decimals)	= ROUND(Town Budget / Total Assessed Value, 4)
Property — Annual Tax Due	= Tax Rate × Assessed Value

Check Your Understanding — 7 Questions

Q1 [Calculation]

Tidal Goods Co. sells an order priced at \$600. The buyer receives a \$100 trade discount. Sales tax is 5.5%. Shipping is \$18. What is the taxable subtotal and the total price?

Answer: Taxable = \$600 – \$100 = \$500. Tax = \$500 × 0.055 = \$27.50. Total = \$500 + \$27.50 + \$18 = **\$545.50**.

Q2 [Concept]

A merchant collects \$4,200 in sales tax over a week. Is that \$4,200 the merchant's revenue? Explain why or why not.

Answer: No. Collected sales tax is a liability held in trust for the state government, not earned revenue. Recording it as revenue overstates income.

Q3 [Calculation]

Register receipts total \$78,000 for the day, inclusive of 7% sales tax. What were the actual pre-tax sales?

Answer: \$78,000 / 1.07 = **\$72,897.20**. The collected tax = \$78,000 – \$72,897.20 = \$5,102.80.

Q4 [Calculation]

Meridian Advisory Group buys office equipment for \$8,500. Sales tax is 6.5% and excise tax is 9%. Calculate the total cost using both methods (separate and combined rate).

Answer: Method 1: ST = \$552.50; ET = \$765. Total = \$9,817.50. Method 2: $\$8,500 \times 1.155 = \$9,817.50$. Both match.

Q5 [Concept]

Why is it incorrect to apply excise tax to the sales-tax-inclusive amount rather than the base cost?

Answer: Both excise and sales tax are calculated from the original base price — not compounded. Applying the second tax to an already-taxed amount overstates the total and inflates cost of goods.

Q6 [Calculation]

A property has a market value of \$450,000. Assessment rate: 35%. Town budget: \$198,000. Total assessed value in town: \$2,600,000. Calculate assessed value, tax rate (to 4 decimal places), and annual tax due.

Answer: Assessed Value = $0.35 \times \$450,000 = \$157,500$. Tax Rate = $\text{ROUND}(\$198,000/\$2,600,000, 4) = 0.0762$. Annual Tax = $0.0762 \times \$157,500 = \$12,001.50$.

Q7 [Strategy]

Meridian Advisory Group is considering a Triple Net (NNN) lease for new office space. Why should they research the municipality's budget trend before signing? What specific financial risk does an NNN lease create?

Answer: Under NNN, the tenant bears property tax, insurance, and maintenance. If the town's budget increases, the tax rate rises and the tenant's costs increase with no cap. An NNN lease converts a fixed rent into variable overhead — the tenant assumes the risk of local government spending.

Key Vocabulary

Term	Definition
Sales Tax	A state-imposed consumption tax on the retail sale of goods and services; collected by merchants on behalf of the government.
Trade Discount	A reduction from the retail price given to a business buyer (e.g., a wholesaler or repeat partner); subtracted before sales tax is applied.

Excise Tax	A targeted tax on specific goods (fuel, alcohol, tobacco, luxury appliances) typically paid upstream by manufacturers or wholesalers and embedded in the price.
Assessed Value	The dollar value assigned to a property for tax purposes: $\text{Assessment Rate} \times \text{Market Value}$.
Assessment Rate	The percentage of a property's market value that a municipality designates as taxable.
Tax Rate (Property)	The rate applied to assessed value to calculate tax due: $\text{Town Budget} \div \text{Total Assessed Value of all property in town}$; rounded to four decimal places.
Nexus	The legal connection (physical presence or economic activity) that requires a business to collect and remit sales tax in a given jurisdiction.
Triple Net Lease (NNN)	A lease where the tenant pays property taxes, insurance, and maintenance in addition to base rent — shifting operating cost risk from landlord to tenant.
Business Personal Property	Movable, tangible operating assets (computers, furniture, machinery) subject to local property tax.
Demand Elasticity	The sensitivity of customer demand to price changes; relevant to excise tax when businesses must raise prices to maintain margin.